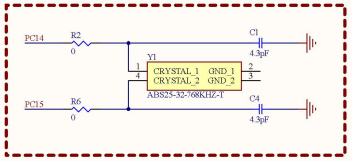
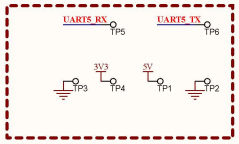


Power configuration

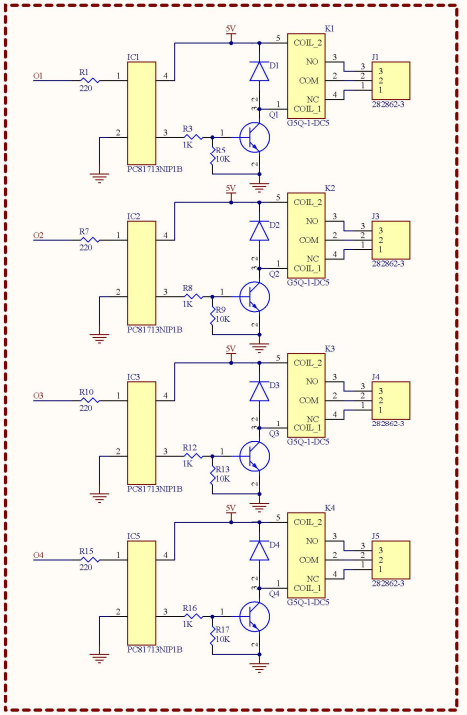
* 220V AC/5V DC to 3V regulator



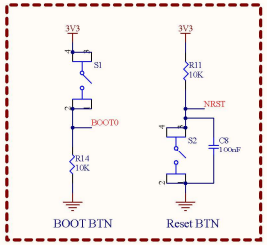
Clock configuration



Testing Points

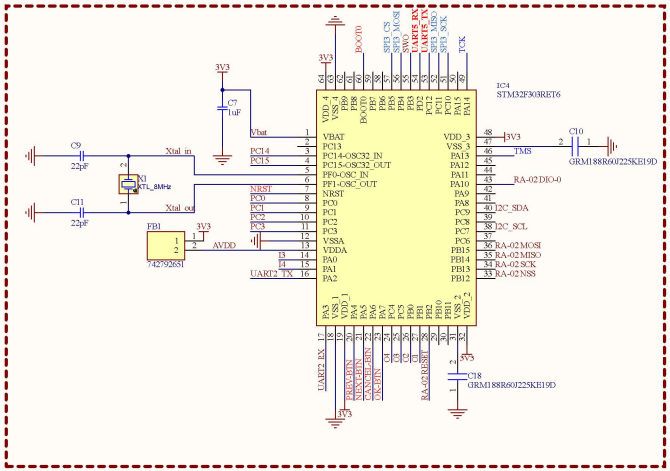


* 4-Relay Output for High-Current Motor Pump Control via Solenoid Valves



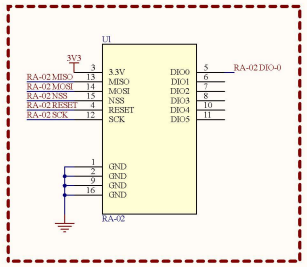
Boot configuration

* Press and hold the **RESET button for 3 seconds** while the firmware is flashing.

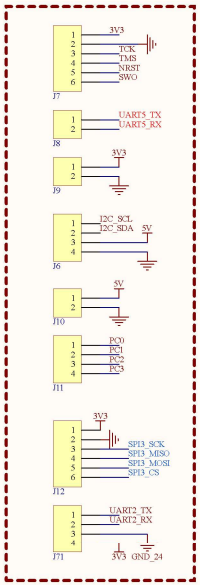


Arm Cortex M4 STM32 f303RE

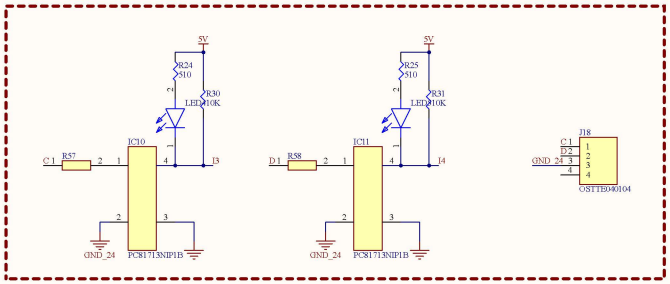
* UART2 TX RX used for flashing and communication with ST-Agro Software



SX1278 - Ra-02 Communication Module

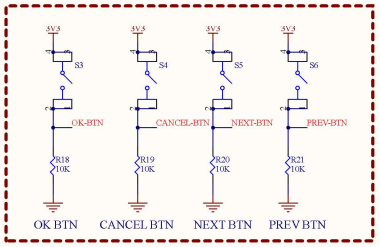


I/O configuration



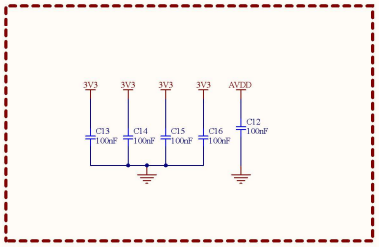
2x Input + Level shifter 24V

* 2× Digital 24V Level Shifter — Supports Industrial Digital Sensors and Switches



Keyboard

* 4-Button Keypad for Output Control and Monitoring Multiple Nodes via I²C LCD (Not Supported in 2.23 Firmware)



Title	ST-Agro Master Gateway Node	
Size: A2	Number	Author: Mazen Daghari — Revision: A.1
Date: 8/23/2026		Sheet of:
File: C:\Users\...\ST-AGRO.SchDoc		Drawn By: